

Lab Report No _7_
Demonstration and practice of Dot Matrix LED
Control and display different patterns.
Embedded Systems



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Lab Task No 3:

Solution:

Brief description (3-5 lines)
Write a program which display the first alphabet of the name of all group members on Dot Matrix LED. Each alphabet stays on 0.5 seconds.
The code
ORG 0000H MOV DPTR, #L_A CALL DISPLAY MOV DPTR, #L_B CALL DISPLAY MOV DPTR, #L_C CALL DISPLAY SJMP \$ DISPLAY: MOV R0, #8 DISPLAY_LOOP: MOVX A, @DPTR MOV P1, A ACALL DELAY INC DPTR DJNZ R0, DISPLAY_LOOP CALL DELAY_500MS RET L_A: DB 0x3C, 0x66, 0xC3, 0xFF, 0xC3, 0xC3, 0xC3, 0x00 L_B: DB 0xFC, 0x66, 0x66, 0xFC, 0x66, 0x66, 0xFC, 0x00 L_C: DB 0x3C, 0x66, 0xC0, 0xC0, 0xC0, 0x66, 0x3C, 0x00 DELAY_500MS: MOV R1, #50 D1: MOV R2, #255 D2: MOV R3, #255 DJNZ R3, D2 DJNZ R2, D1

```
DJNZ R1, DELAY_500MS
RET
DELAY:
MOV R4, #100
D3: DJNZ R4, D3
RET
END
```

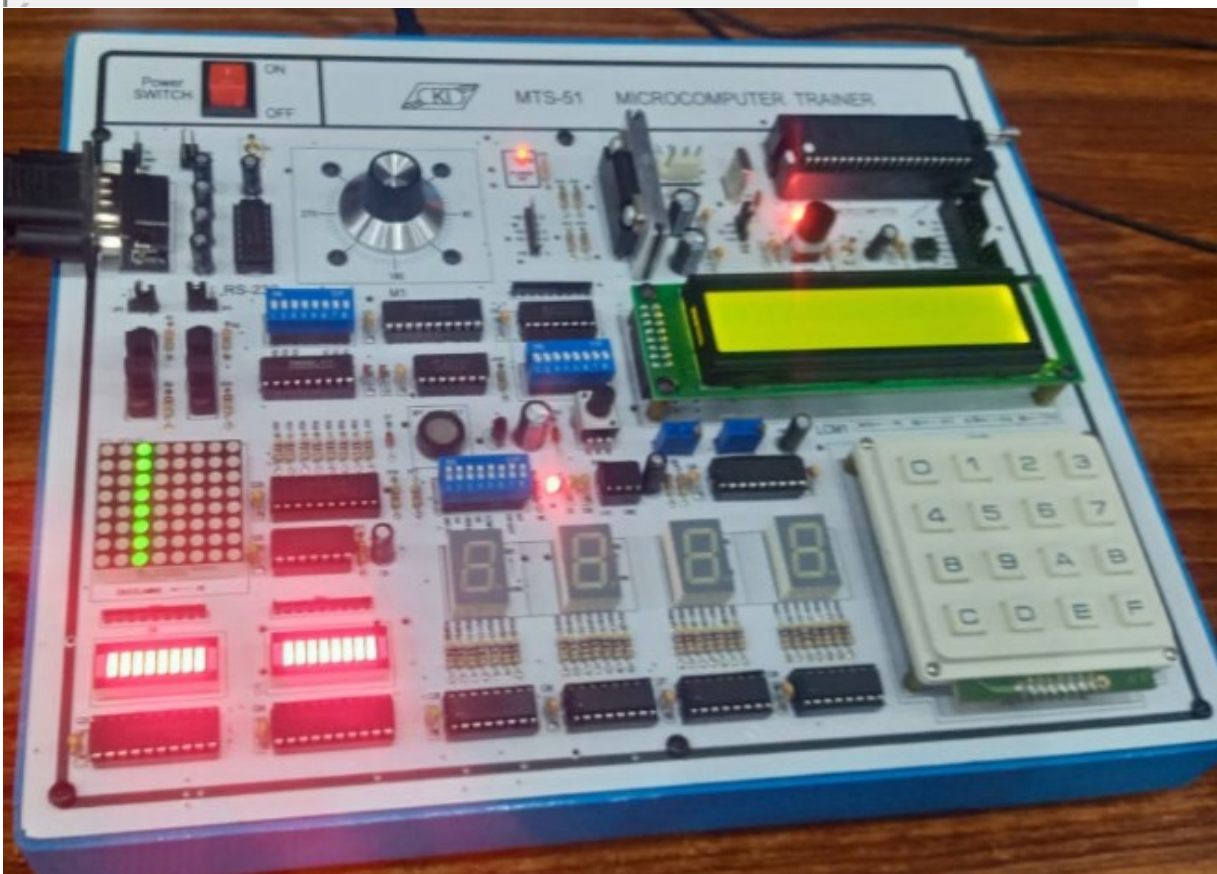
The results (Screenshot)

Build Output

```
Program Size: data=9.0 xdata=0 code=72
".\Objects\dotmartixledaplahabet" - 0 Error(s), 3 Warning(s).
Build Time Elapsed: 00:00:00
```

Build Output

```
creating hex file from ".\Objects\dotmartixledaplahabet"...
".\Objects\dotmartixledaplahabet" - 0 Error(s), 3 Warning(s).
Build Time Elapsed: 00:00:01
```



Conclusion

In conclusion, the Dot Matrix LED control project provided practical insights into programming microcontrollers for display purposes. It emphasized the importance of precision in timing and memory management, especially in resource-constrained environments. The project demonstrated the versatility and dynamic nature of Dot Matrix displays for real-time applications, underscoring their potential in visual communication within embedded projects. This exercise underscored the potential of Dot Matrix displays in embedded projects.